COMP 3920 Database Systems 1

Assignment 2

Introduction:

In class we've been designing and writing queries for a Chat Messaging System and now we get to built it!

Your site will be written in Node.js and will interact with a MySQL database.

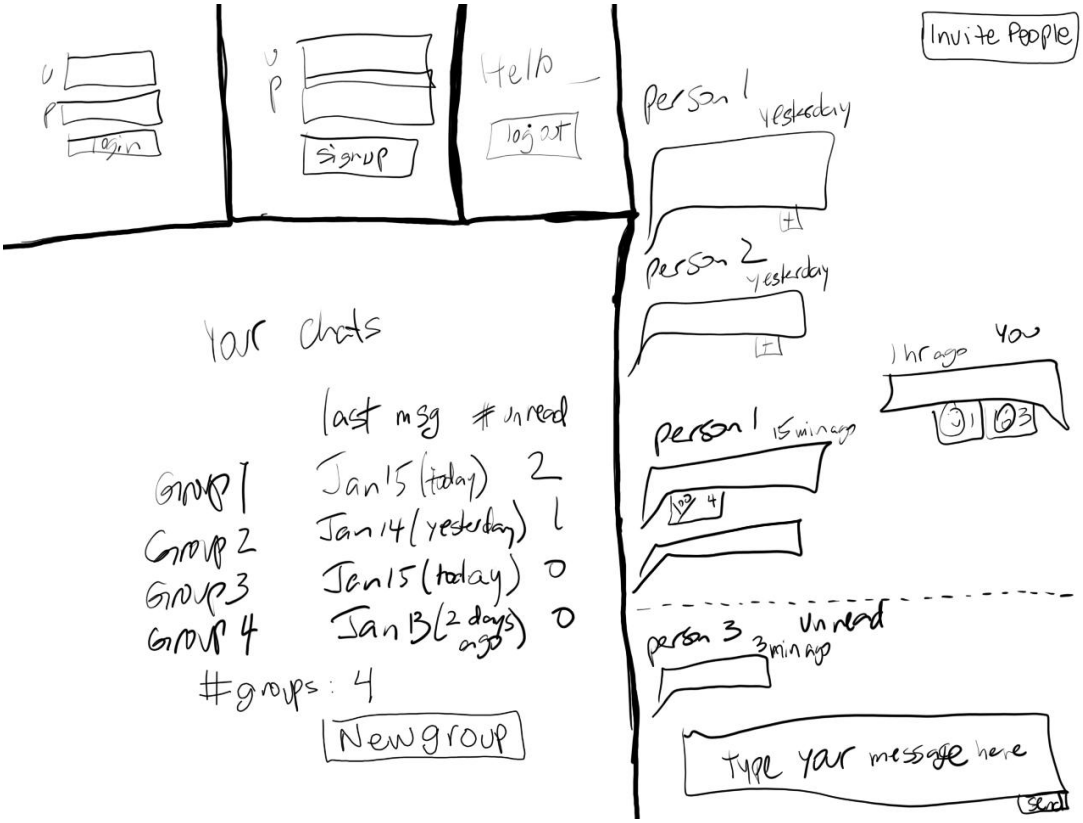
Your site will be hosted on a service like Render. In order to have a publicly accessible database that our hosted site can use, we can use a MySQL Host service like Aiven.io or FreeDB.tech.

Requirements:

Required functionality and best practices:

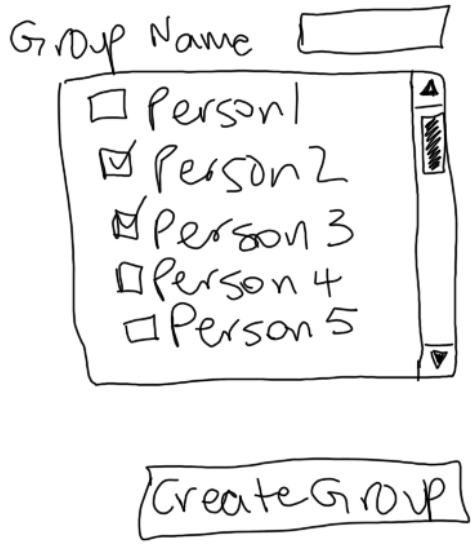
- Authentication (login with hashed passwords, password validation >= 10 characters with upper/lower, numbers, symbols)
- Must be logged in to read and send messages.
- Authorization (you can only read and send messages in your chat groups, any attempt to access anyone else's results in a 400 error)
- Sessions (Cookies in an encrypted MongoDB)
- Hosted on Render (or other Hosting service)
- Good database design hosted on Aiven.io or FreeDB.tech

Professional application. Bug free (well tested). Production quality.



There are a number of different pages mocked up in the diagram above:

1. A log in page.
2. A sign up page.
3. Once you are logged in, at the top of each page, it should show the username of the logged in user and a log out button that will log the user out (destroys the session).
4. A page to show all of the chat groups that you are in:
    a. Show a list of groups that you are in.
    b. Show the date of the most recent message in this group.
    c. Show the number of unread messages (messages that you haven't seen). Clear unread messages once you leave the page.
    d. A total of the number groups that you are in.
    e. A button to create a new group.
5. Clicking on a group chat will take you to a list of the messages within that group:
    a. A button to invite more people to the group (clicking on this button should also show the currently members of this group).
    b. Show all the messages in this chat group in "sent" order (newest on the bottom).
    c. Show the date each message was sent.
    d. Show the associated emoji reactions to each message and a button to add more emojis.
    e. A text field and a button to send a new message.



The above mockup shows a "Create Group" page where you can give your group a name and select from a list of users to add to the group.

Marking Guide:

Criteria	**Marks**
Appropriate Sign up, Log in, Authentication, Authorization, Log out `.env` file for secrets, BCrypted Passwords and MongoDB Encrypted Sessions	20 marks
Ability to see group chats that you are in along with last message date and number of unread messages	10 marks
Ability to create new group chat and invite people	5 marks
Show all messages within a group chat with date sent with most recent on the bottom	5 marks
Ability to see and add reaction emojis to individual messages	10 marks
Ability to see and clear unread messages	5 marks
Ability to send a new message	5 marks
Subtotal:	**60 marks**
You will receive a mark of 0 if your code does not compile or has too many errors to run!	

Submission Requirements:

Submission:	**File name:**
A txt file containing the Hosted app URL. ex: `http://xrb198fpq1a.us11.renderapp.com/`	Hosted_app_URL.txt
A zip file containing **all the source code** required to run your Node.js application. Do **NOT** include the `/node_modules/` or the `/.git/` folder	Chat_Messaging_site_code.zip
A txt file containing the URL of your github repo (Also, make sure to add me as a collaborator: GitHub User: @greencodecomments)	GitHub_Repo_URL.txt